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I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Items we have Discussed

During Week 3, the project progressed significantly in both hardware and software domains. The primary focus of this week was to initiate the physical setup of the drone system by acquiring and understanding essential drone components, while simultaneously enhancing the AI capability of the system by integrating a more refined model trained on Google Colab. This week marks a crucial step toward building a complete cyber-physical system that combines drone hardware, cloud-based AI processing, and mobile application interaction.

1.1 Hardware Component Acquisition and Understanding

In this week, essential drone components were procured to begin the hardware development phase of the AgriDrone system. The components included a drone frame (composed of carbon fiber and plastic), four pairs of propellers, four brushless motors rated at 1400W, four Electronic Speed Controllers (ESCs), a LiPo battery (2200 mAh), and a compatible battery charger. Each component was studied to understand its role within the drone system. The frame serves as the structural base, providing support and alignment for all components. The brushless motors generate thrust, while the propellers convert rotational motion into lift. ESCs are responsible for controlling motor speed based on signals from the flight controller. The LiPo battery acts as the primary power source, and its capacity determines flight duration and power availability.

1.2 Hardware System Planning

Although full assembly was not completed during this week, a clear understanding of component interconnection was developed. The conceptual workflow of the drone system was established, where the battery supplies power to ESCs, ESCs drive the motors, and the motors rotate propellers to enable flight. Safety considerations such as proper battery handling, ESC calibration, and motor alignment were also studied to ensure safe implementation in the upcoming stages.

1.3 AI Model Enhancement Using Google Colab

In parallel with hardware development, the AI component of the system was significantly improved. A more refined dataset for rice and wheat disease detection, trained using Google Colab, was integrated into the project. The notebook (agridrone_train.ipynb) was analyzed to understand the training process, which utilized the Ultralytics YOLO framework. The trained model (best.pt) was obtained from the Colab environment, representing a more accurate and generalized model compared to the locally trained version.

1.4 Integration of Colab-Trained Model with Backend

The trained model from Google Colab was successfully integrated into the FastAPI backend. The API was updated to load the new model and process incoming image

requests using this improved AI model. The system now supports prediction for multiple crops, specifically rice and wheat, based on the trained datasets. The backend processes images received from the Flutter application and returns disease predictions along with confidence scores.

1.5 System Integration and Testing

The complete system workflow was tested, integrating the Flutter mobile application, FastAPI backend, and AI model. The application successfully sends plant images to the backend, which processes them using the trained model and returns predictions. The system also stores prediction results in Firebase, enabling real-time data updates in the mobile application. Initial testing showed improved performance compared to the previous model, although some prediction inconsistencies were observed.

2. Problems we Faced

During this week, several challenges were encountered across both software and hardware components. One major issue was inaccurate disease prediction results when testing with real-world images. This occurred due to differences between training dataset conditions and actual field images, such as lighting variation, background noise, and image clarity. Additionally, handling detection-based AI models posed difficulties, as their output format differs from classification models, requiring adjustments in backend logic. On the hardware side, understanding the safe usage of LiPo batteries and proper configuration of ESCs required careful study, as improper handling could lead to system damage or safety risks.

3. Problems we Mitigated

To address these challenges, several solutions were implemented. A more refined and well-trained model from Google Colab was integrated to improve prediction accuracy. The FastAPI backend was modified to correctly interpret detection model outputs by selecting the most confident prediction. For improving system reliability, additional testing was performed using different image conditions. On the hardware side, extensive research and documentation review were conducted to understand safe battery handling, ESC calibration, and motor configuration, ensuring readiness for future assembly and testing.

4. Achievement of the Week

The key achievement of this week was the successful advancement of the project from a software-focused prototype to a more complete system integrating both hardware and improved AI capabilities. A major milestone was the acquisition and understanding of essential drone components, including the frame, motors, ESCs, propellers, and power system, which marks the beginning of the physical implementation phase of the AgriDrone.

In addition, a significant improvement was achieved in the AI component by integrating a more refined model trained on Google Colab using a larger and better-structured dataset for rice and wheat disease detection. This enhanced model improved the overall prediction capability of the system and enabled support for multi-crop analysis.

Furthermore, the integration of this trained model with the FastAPI backend and Flutter application established a stronger end-to-end pipeline, where real-time image input from the mobile app is processed using an advanced AI model and results are displayed and stored effectively. This week therefore represents a critical transition toward building a fully functional AI-powered agricultural drone system.

5. Tasks for next Week

In the upcoming week, the focus will be on advancing both hardware assembly and system refinement. The primary task will be assembling the drone components, including mounting motors on the frame, attaching propellers, and connecting ESCs to the power system. Proper wiring, calibration, and safety precautions for handling the LiPo battery will be ensured. Additionally, initial integration of the flight controller will be explored, along with basic configuration and synchronization of motors and ESCs to achieve stable operation.

On the software side, efforts will be made to improve the AI model by analyzing incorrect predictions and refining the dataset. Further testing will be conducted using real-world images to evaluate and enhance model performance. The Flutter application will also be improved by adding features such as crop selection and better visualization of prediction results. Finally, initial steps will be taken to integrate hardware and software components, moving closer to a fully functional prototype.

4.1 Quantitative Progress Metrics

The following measurable achievements were accomplished this week:

- Hardware Components: 100% of core drone components acquired and documented
- AI Model Performance: Achieved mAP@0.5 of 56.6% on rice disease validation set
- Dataset Expansion: Integrated 2,500+ annotated images for rice and wheat diseases
- System Integration: Established complete data pipeline (Flutter → Backend → AI → Firebase)
- Testing Coverage: Validated system with 50+ test images across different conditions
- Model Optimization: Reduced inference time to 50-80ms per image
- Multi-Crop Support: Successfully extended system from single-crop (rice) to dual-crop (rice + wheat)

These metrics demonstrate substantial progress toward a production-ready agricultural monitoring system.